

(30)

ECE 4644 / 5610 Satellite Communications

Homework Assignment #7 – Complete SOLUTIONS

Problem #1

41 Explain in general terms what is meant in communication systems by the terms *baseband signal*, *RF carrier*, and *modulation type*.

Specify what is in play for each of these italicized terms for the over-the-air broadcast station WUTF 89.1

Problem #2

The term *latency* in communications refers to the delay in signal reception due to the finite speed of propagation of the signal. For satellite communications we are interested in round trip (or two-way) latency which encompasses the delays due to propagation in space from the transmitter to the satellite and then to the receiving Earth station. Processing of the signal also contributes to latency.

a) Estimate the round-trip latency due to propagation in space for these satcom scenarios (would be useful to show the calculation once and then arrange results in a Table):

- 51
- i) a satellite in LEO orbit at 500 km
 - ii) a satellite in GEO orbit at 35,786 km
 - iii) the Artemis s/c at the height of its HEO orbit at 70,000 km
 - iv) the Artemis s/c at its greatest distance from Earth at 404,500 km (setting a record)

b) Comment on the impact of propagation latency on voice communications, text communications, and data transfer. That is, for particular applications assess propagation latency as a concern (or not). Be quantitative and write no more than half a page (typed).

Problem #3

A baseband signal has a one-sided power spectrum that is shaped like a symmetric triangle in frequency with peak power at 3 kHz, zero power at 0 Hz, and zero power at 6 kHz and above. The baseband is sampled using infinitely short pulses.

a) Sketch the power spectrum as $V(f)$ and explain what minimum sampling rate, f_s , is needed to reproduce the baseband from discrete samples.

61 b) Sketch the pattern of sampling pulses in time using the separation that corresponds to the minimum sampling rate.

c) Sketch the spectrum of the signal after sampling of baseband using the pulses. (Note: the two-sided power spectrum of $V(f)$ extends to negative frequencies – via reflection through the $f = 0$ axis – which is usually not plotted. After sampling, however, that content will appear at positive frequencies and should be drawn.)

d) Explain that we can recover the content of the baseband signal by applying a low-pass filter to the sampled signal.

Problem #4

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Artemis II is equipped with an optical system known as O2O for high-speed communications with Earth, in particular for streaming video and sending high-res images. The system is based on a laser that operates at an infrared wavelength of $1.55 \mu\text{m}$.

- Determine the operating frequency of the system and contrast it with the S-band that is designated for space research and deep-space operation
- The system uses a telescope device that was developed at MIT Lincoln Laboratory. It has an aperture of 4 inches. Assuming that we can apply the same equations, estimate the beamwidth of the device
- Estimate the length of the arc of Earth's surface that would be illuminated by the device when it is at the distance of the moon (no lengthy calculations required). Comment on the implications for operations.
- Speculate on why the O2O was set up to operate on an infrared frequency rather than an optical frequency

Problem #5

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✓
A communication system involves the conversion of analog input to digital signal. Sketch a block diagram that depicts an analog voltage input signal undergoing sampling and then conversion to digital signal and finally conversion to a stream of bits. Explain with brief design notes the function of each element in the block diagram. Indicate where the signal conforms to Pulse Amplitude Modulation and where it conforms to Pulse Code Modulation, and explain the meaning of these terms. (You may use figures from the slides.)

Problem #6

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✓
The standard word used in a digital telephone channel consists of 8 bits and has a sampling rate of 8 kHz. The nominal voltage range is $\pm 0.775 \text{ V}$. Assume uniform quantization over this full voltage range.

- Explain why 8 kHz is a reasonable choice for sampling baseband, which in this case consists of the human voice.
- Determine the bit rate of the digital telephone channel.
- Find the number of quantization levels, the step size of the quantization, and the maximum magnitude of quantization error.

Σ = 30

Problem #1.

Baseband Signal-The broadband signal is the original, lower frequency the signal is generated at without a carrier and before any modulations – this signal can be voice, video, or data transmissions. For digital signals, the baseband can be thought of as a transmitted series of logical 1s and 0s carried by rectangular pulses made up of positive and negative voltage changes. ✓

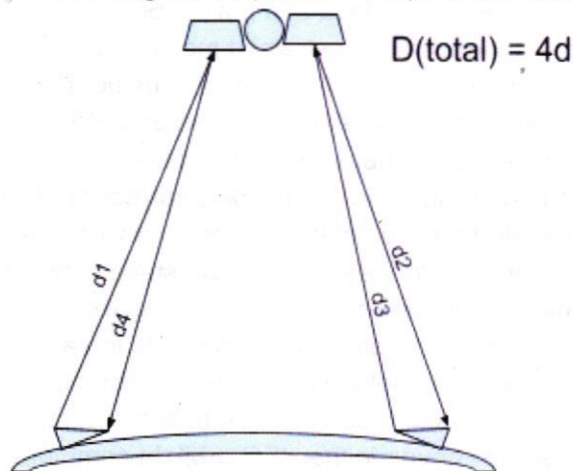
RF Carrier- The carrier wave performs its namesake and carries the baseband signal to its destination. An RF carrier signal usually has a higher frequency than the baseband signal. A Baseband signal is impressed onto the carrier signal using modulation. The higher frequencies of the carrier waves allow multiple basebands to be transmitted without interference and for more accessibility in terms of receiving antenna requirements. ✓

Modulation type- The way a baseband signal is impressed onto a carrier signal, it can modulate any of the signal parameters (frequency, amplitude or phase) which determines the type of modulation. Modulation can be split into digital and analog types where digital uses 0s and 1s and analog uses continuous waveforms. Types of analog modulation include AM and FM and types of digital modulation include BPSK, QPSK, QAM and APSK. ✓

For WVTF the baseband signal would be the original signal of the broadcast before any modulations in the frequency the audio was recorded in. The RF carrier wave is how the signal was transported, in this case using the frequency of station 89.1 MHz. The modulation type given the station number is FM, frequency modulation – this means that the frequency was modified to impress the baseband signal onto the carrier wave. ✓

Problem #2.

a) Where round trip is defined as station 1 sending a signal, station 2 receiving it, retransmitting to station 1 and station 1 receiving the response – 4 trips to the satellite total.



Total latency = propagation delay + processing delay. However, the following calculations focus on propagation delay. We know $d = c \cdot \Delta t$ or $\Delta t = \frac{d}{c}$. Using the speed of light constant in km, $c = 299,792.458$ km/s.

Scenario	Equation	Propagation Delay (s)
LEO orbit @ 500 km	$\Delta t = \frac{4 \cdot 500}{299,792.458}$	0.0067 seconds
GEO orbit @ 35,786 km	$\Delta t = \frac{4 \cdot 35,786}{299,792.458}$	0.4775 seconds
Artemis s/c in HEO orbit @ 70,000 km	$\Delta t = \frac{4 \cdot 70,000}{299,792.458}$	0.9340 seconds
Artemis s/c @ 404,500 km	$\Delta t = \frac{4 \cdot 404,500}{299,792.458}$	5.3971 seconds

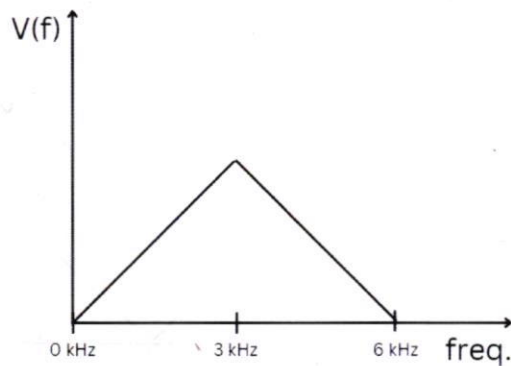
b) When considering the impact of propagation latency, I first consider the natural cadence of the communication types. The biggest concern for latency would be for voice as the natural response time for conversation is fairly quick, under 1 second. Voice communication resembles in person human interaction the most and therefore requires a response time similar to an in person conversation and would benefit the most from a low propagation delay. I would put the delay tolerance of a voice connection at a max of .5 seconds. The least concern for latency would be text communication. I am not a prompt texter, so this definitely creates a bias on my choice of least concern for latency. I would not notice a minute delay much less a second delay when using text communication. I believe this is a socially accepted stance for text communication so I would set the delay tolerance at a max of 60 seconds. This tolerance still allows time for 2-Factor and other more time constrained messages. For data transfer propagation delays, depending on how the transferred data is being utilized the propagation delay could be more noticeably impactful. For example, if the data is being used for a data transfer or data ingestion into a database where time is not a sensitivity, there is room for propagation delay as long as the receiving end's network is correctly designed to expect the delay between packets or acknowledgments, toleration delay max of 5 seconds would be fine. I could see propagation delay playing a larger effect with high stake services that are expected to behave at "real-time". Stock market databases, security alarming real time systems, would be very noticeable with a fraction of a second delay. Given the table in part one, text messages would be the most tolerable for the record setting Artemis crew distance, data for photos and updates would also be tolerable – we would receive them eventually, but we wouldn't want to run any stock exchanges from Artemis even in the HEO orbit. In communications with the crew, the voice conversations with the ~5 second delays were awkward.

Problem #3.

Peak Power @ 3 kHz

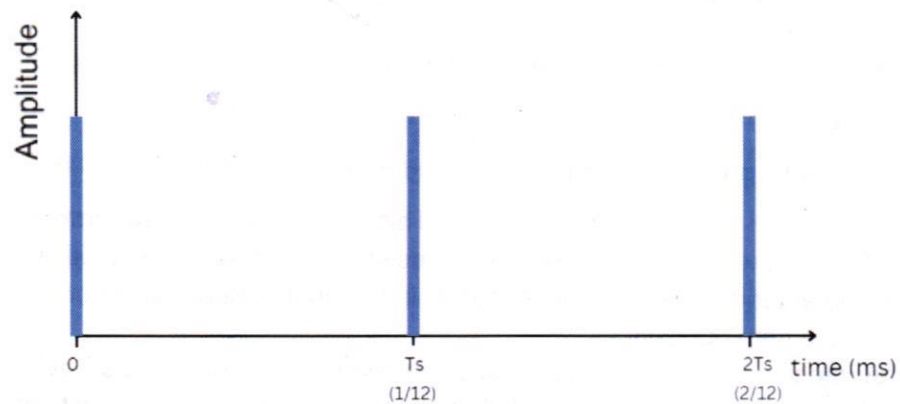
Zero Power @ 0, > 6 kHz

a)

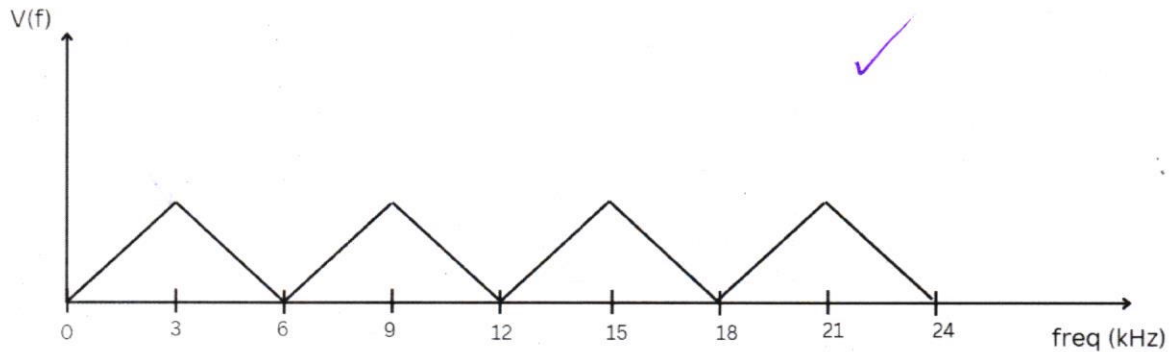


We know that $f_{\text{samp}} > 2 \cdot f_{\text{max}}$ from the Nyquist Sampling Theory. From the sketch, we know that max frequency is 6 kHz. After plugging that into the theory equation, we know that the minimum sampling frequency would be 12 kHz.

b) To find the the timing of the sampling pulses we will use the equation $T_s = \frac{1}{f_{\text{samp}}}$



c) While we can't combine the time and frequency domains directly, we can do a fourier transform from the time domain to the frequency domain. This gives us pulses at f_{samp} , $2f_{\text{samp}}$, $3f_{\text{samp}}$, etc. Therefore the signal spectrum would include the original signal and its mirror repeated at every interval of f_{samp} as shown in the figure below.



d) Applying the baseband sampling repeated the signal across higher frequencies but the original signal remains intact. We can recover the content of the original signal using a low-pass filter with a cutoff at f_{max} . This ensures the original baseband signal passes the filter but filters out the higher frequency replications caused by sampling.

Problem #4.

$= 6 \text{ KHz}$

a) First, I need to convert the infrared wavelength to cm to use the magic equation from class. $1.55 \mu\text{m} = 0.000155 \text{ cm}$. Where $f[\text{GHz}] \cdot \lambda[\text{cm}] = 30$, $f = 193,548 \text{ GHz}$. In comparison to S-Band (2-4 GHz), the wavelength of the infrared is almost 5 orders of magnitude larger than low S-Band frequencies.

b) First, I will convert inches to cm for the aperture, $D = 10.16 \text{ cm}$. We will use the beamwidth estimation formula $\theta \approx \frac{75\lambda}{D}$, $\theta \approx \frac{75(0.000155)}{10.16}$, $\theta \approx 0.0011$.

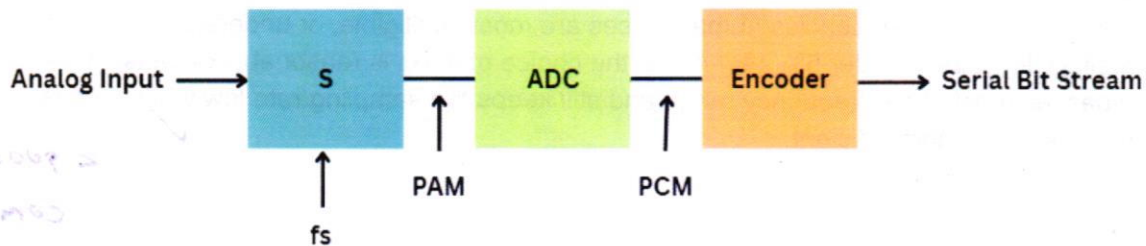
c) First, convert part b into radians = 0.000019 and find $R_{\text{moon}} = 384,400 \text{ km}$, Using the equation $l = \theta \cdot r$, $l = 0.000019 \cdot 384,400 = 7.3 \text{ km}$. This is a very small area, orders of magnitude smaller than RF, that would require extremely precisely placed and continuously calculated ground sites and antennas. The smallest change in the orbit or beam direction would make it difficult to continue communications.

d) Optical frequencies would result in a smaller wavelength and therefore an even smaller beamwidth. Given how challenging the infrared beamwidth size seems, I don't imagine going smaller would be feasible or worth the technical demand it would require.

$< \text{also atm. effects}$

Problem #5:

Using our course slides, the following figure is a block diagram of converting an analog signal to a serial bit stream.



Block S: A sampler is used to take samples from the signal. It is important to take samples at a high enough sampling rate (twice the max signal frequency)

f_s : The sampling frequency used for the sampler

PAM: Pulse Amplitude Modulation (PAM) is the output from the sampling process. These samples can still have any value but occur in pulses at intervals of f_s .

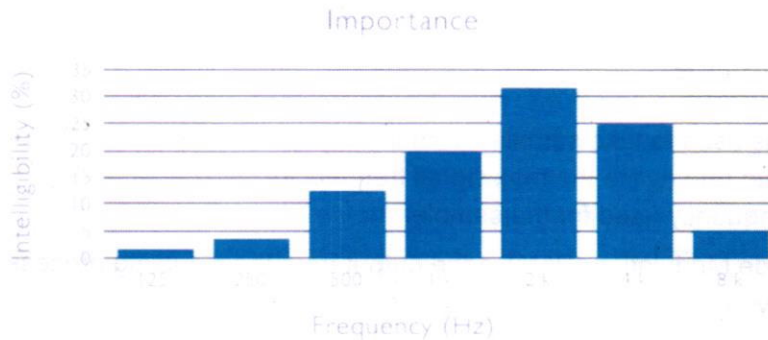
Block ADC: ADC achieves matching the corresponding digital words to the voltage samples and can be characterized by speed and number of bits in each word. This outputs the word that has a matching voltage closest to the input signal.

PCM: Pulse Code Modulation (PCM) are the digital words that resulted from the digitally encoded samples with quantization in the ADC. At this point, the encoded samples can only have discrete values determined by quantization, essentially the pulses (PAM) have been rounded to fit predetermined values (PCM).

Block Encoder: The encoder takes the quantization values and assigns them binary values. The number of values is dependent on the bit rate or how many bits per sample. The output of this block is the serialized bitstream.

Problem #6:

a) Using the Nyquist Sampling Theory, we know that the sampling frequency should be at least twice the max signal frequency. Given a 8kHz sampling baseband, we have a max signal frequency of 4kHz. After researching the frequency of human voice, I found the following graph that highlights at what frequencies human voices are most intelligible, or understandable. Given that intelligibility falls to under 5% after 4kHz, the choice of 4kHz is reasonable because it encompasses most of the frequency range and still keeps the sampling rate low which keeps the overall system more efficient.



✓
← good comment

[[chart source](#) / data source: N.R. French & J.C. Steinberg: Factors governing the intelligibility of speech sounds. JASA vol. 19, No 1, 1947]

b) bit rate = $f_s \times N = 8\text{kHz} \times 8 \text{ bits} = 64 \text{ kbps}$.

c) The number of quantization levels can be calculated by $2^8 = 256$. Assuming equally spaced out steps, the step size of the quantization is $\frac{.775 - (-.775)}{256} = 0.006 \text{ V}$. The maximum error of quantization is when the PAM signal is right in between two steps, this can be shown by $\frac{\Delta V}{2}$ or .003 V.